

Syllabus

Spring semester 2026

Course Title	Fundamentals of Analytical Chemistry	Course Code	CB4103	Credit	3	Classroom	5605, 5606
		Type	elective	Hours	3		3304, 2503
Teacher	Name	Class no.	Office	Office hours	Contact	E-mail	
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1. Course Description

As chemistry is the science of molecule, chemistry studies the component and structure of materials, identify the properties of chemicals, and study the reaction from reactants to products with energy transfer. This course provides an introductory survey on traditional methods (titration, gravimetric, and colorimetric method) and modern methods (instrumental methods including spectroscopy, electrochemical methods, and chromatography) as fundamental level. Develop the intellectual skills to integrate theory and practice in student-centered class with cooperative learning. Understand the qualitative and quantitative analysis with personal and team projects. Learn the process of data analysis like the statistical treatment of data and literature survey and validation of analytical methods.

Analytical chemistry can be used to various fields as basic tools, this course gives a chance to connect to other sciences and environmental fields. Students take experiences on chemical analysis from the understanding the fundamental theory of instrument to apply the knowledge for the field using the KSA instrument and tools through the projects.

- 1) Intensify the fundamental understanding of analytical techniques
- 2) Accustom the instrument and practice the application methods to solve the real problems
- 3) Choose appropriate information and improve scientific communication skills
- 4) Rigorous interpretation and statistical analysis of data

Contents and Guideline

The prerequisites courses are General Chemistry1 and General Chemistry Lab1. This course provides the fundamental understanding and practices of chemical equilibrium, electrochemistry, spectroscopy, and separation science. The course also includes qualitative and quantitative experiment and projects using classical and instrumental methods.

- 1) Lab safety and chemical measurement tools and apparatus of lab
- 2) Experimental error and Statistics
- 3) Acid-base
- 4) Electrochemistry
- 5) Complex and precipitation
- 6) Titration
- 7) Extraction and chromatography

- 8) Spectroscopy
- 9) New analytical instruments and methods
- 10) Individual project using instrument with operation, demonstration, presentation, and discussion
- 11) Application to real problems (team project, improve science communication skill, choose appropriate information, Experience of reproducible data acquisition following the analytical methods, Rigorous interpretation and statistical analysis of data)

2. Core Target Competencies

Subject Core Competencies						
analytical and creative thinking	problem-solving ability	science research ability	digital and quantitative literacy	leadership and teamwork	complex communication	
* Blank fields <u>to</u>can be filled in accordingly <u>ly</u>-to the situation. * Fill in the blanks <u>here</u> with descriptive words.						

3. Textbooks and References

- A. Textbook: Quantitative Chemical Analysis 9th ed. Daniel C. Harris. W. H. Freeman and Company
- B. References
 - 1) Fundamentals of Analytical Chemistry, 8th ed., D. A. Skoog etc, Thomson and Brooks/cole
 - 2) Analytical Chemistry, David S. Hage and James D. Carr, Pearson Education
 - 3) Practical Skills in Chemistry, 3rd ed. John Dean etc, Pearson
 - 4) Quantitative Chemical Analysis 10th ed. Daniel C. Harris. W. H. Freeman and Company
 - 5) Encyclopedia of analytical chemistry: applications, theory and instrumentation, R. A. Meyers, Chichester : John Wiley & Sons. Ltd , 2000

4. Class Method

- A. Lab reports and projects must be submitted by the deadline, or there will be a penalty on the scores.
 - B. Teaching and Learning
 - 1) Student-centered
 - 2) Inquiry-based learning and problem-solving (POGIL)
 - 3) Self-directed learning with individual learning
 - 4) Cooperated learning and improving the communication through team project
 - 5) Rigorous interpretation and statistical analysis of data
 - C. Class hour: some labs and projects can be done on weekend
- ✱ Students form teams and conduct classes through team problem-solving activities. Students think about learning objectives through a discussion process and learn core scientific concepts and scientific knowledge while solving given problems through collaboration.
- Autonomously form small groups and discuss learning objectives.
 - Collaborate in small group activities to solve problems together and present your understanding to your peers.
 - Share the class materials you solved together and write reflective essays based on what you understood.

5. Evaluation

A. Evaluation Criteria

Lab activity	Presentation and project	Quiz	Mid-term exam	Final exam	Attendance and Attitude	Sum
10	35	5	20	20	10	100

B. Evaluation Method

- (1) Grading Scale: 100-90: A+, 90-85: A0, 85-80: A-, 80-75: B+, 75-70: B0, 70-65: B-, 65-60: C+, 60-55: C0, 55-50: C-, 50-45: D+, 45-40: D0, 40-35: D-, 35-: F.
- (2) Simple lab practice is practice of basic lab techniques in less than one and half-hour
- digital and quantitative literacy (SOP, data analysis with statistics) 50%
 - leadership and teamwork (team work) 30%
 - complex communication (report, presentation) 20%
- (3) Quiz is the statistical data analysis with Excel spreadsheet.
- digital and quantitative literacy 80%
 - problem-solving ability 10%
 - science research ability 10%
- (4) Individual project is to understand and accustom the modern instrument by direct use. It requires the operation demonstration to class, teaching the theory and data analysis to class, and discussion (content is based on text. require the discussion during the preparation of presentation and the training of operation instrument).
- science research ability (instrument operation) 40%
 - digital and quantitative literacy (SOP, report, manual, survey) 30%
 - complex communication (presentation, demonstration) 20%
- (4) Team project (3 student in one team) is practice the whole process of analytical chemistry with SOP. It also requires students to discuss the progress with instructor regularly (proposal, lab note, lab report, logbook of instrument having operation days, statistical methods for data, conclusion and discussion, presentation and final reports). The presentation and final reports contain the content in the text.
- analytical and creative thinking (theory, reference) 10%
 - problem-solving ability (make use objects, chemical work) 10%
 - science research ability (preparation, analysis, using instrument) 30%
 - digital and quantitative literacy (SOP, report, survey, statistical analysis) 10%
 - leadership and teamwork (team work, collaboration) 30%
 - complex communication (presentation, discussion) 10%

서식 있음: 눈금에 맞춤, 탭: 30.08 글자, 왼쪽 + 8 글자(없음) + 12 글자 + 16 글자 + 20 글자 + 24 글자 + 28 글자 + 32 글자 + 36 글자 + 40 글자 + 44 글자 + 48 글자 + 52 글자 + 56 글자 + 60 글자 + 64 글자 + 68 글자 + 72 글자 + 76 글자 + 80 글자 + 84 글자 + 88 글자 + 92 글자 + 96 글자 + 100 글자 + 104 글자 + 108 글자 + 112 글자 + 116 글자 + 120 글자 + 124 글자 + 128 글자

6. Criteria for Awarding Credits

- Pass: In the case of receiving a **score of 80 or more** out of 100 according to the criteria of the evaluation method.
- Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal’s Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.

- Attendance for at least three-quarters of the total class hours is required to earn credit.
- For Pass/Fail courses, if the instructor determines that a specific evaluation element is insufficient and requires a separate evaluation, a separate evaluation may be conducted.
- Calculation of the basis score for awarding credit to absentees :
- If a basis score cannot be calculated, the grading method : committee of chemistry and biology
- ※ Refer to the "Academic Performance Management Regulations" for the percentage of credit points applied.

Final letter grade will be awarded based on an absolute evaluation of the AP grading criteria and the total number of points received by the student.

결시자 인정점 부여는 다음과 같이 계산한다.

1. 시험을 치른 시험에서 학생의 점수 순위를 구한다. 2. 결석한 시험의 해당 순위의 점수를 찾아서 부여한다. 3. 공결이면 100%인정, 병결이면 80%인정한다. 1. Determine the student's score ranking in the exam they took. 2. Find the score corresponding to that ranking for the missed exam and assign it. 3. 100% recognition for official absence, 80% recognition for sick absence.

7. Weekly Plans

Week (period)	Topic and Contents	(Notes)
Week 1 2.23.~3.1.	Introduction, Overview of Analytical Chemistry	3.2. Substitute Holiday
Core Competencies	♦ digital and quantitative literacy, complex communication	
Week 2 3.2.~3.8.	Tools, Experimental Error, Statistics	
Core Competencies	♦ analytical and creative thinking, science research ability	
Week 3 3.9.~3.15.	Statistics, Qualitative Assurance and Calibration method	
Core Competencies	♦ problem-solving ability, digital and quantitative literacy,	

	leadership and teamwork, complex communication	
Week 4 3.16.~3.22.	Qualitative Assurance and Calibration method	
Core Competencies	♦ digital and quantitative literacy, problem-solving ability, science research ability	
Week 5 3.23.~3.29.	Titration, Gravimetric and Combustion analysis	
Core Competencies	♦ problem-solving ability, complex communication, leadership and teamwork, digital and quantitative literacy	
Week 6 3.30.~4.5.	Acid-Base equilibrium, Buffer	
Core Competencies	♦ digital and quantitative literacy, leadership and teamwork, complex communication, science research ability	
Week 7 4.6.~4.12.	Polyprotic titration and EDTA titration,	
Core Competencies	♦ problem-solving ability, complex communication, leadership and teamwork, digital and quantitative literacy	
Week 8 4.13.~4.19.	Midterm Exams (4.14.~4.17.)	4.13. Regular Classes

Week 9 4.20.~4.26.	Electrochemical Analysis	
Core Competencies	problem-solving ability, complex communication, leadership and teamwork, digital and quantitative literacy	
Week 10 4.27.~5.3.	Spectroscopy	
Core Competencies	♦ problem-solving ability, complex communication, leadership and teamwork, digital and quantitative literacy, science research ability, analytical and creative thinking	
Week 11 5.4.~5.10.	Presentation of project 1	5.5. Children's Day
Core Competencies	♦ problem-solving ability, complex communication, leadership and teamwork, digital and quantitative literacy, science research ability, analytical and creative thinking	

Week 12 5.11.~5.17.	Mass Spectrometry	
Core Competencies	♦ problem-solving ability, complex communication, leadership and teamwork, digital and quantitative literacy, science research ability, analytical and creative thinking	
Week 13 5.18.~5.24.	Chemical Separation, GC, LC	
Core Competencies	♦ problem-solving ability, complex communication, leadership and teamwork, digital and quantitative literacy, science research ability, analytical and creative thinking	
Week 14 5.25.~5.31.	Presentation of project 2	5.25.
Core Competencies	♦ problem-solving ability, complex communication, leadership and teamwork, digital and quantitative literacy, science research ability, analytical and creative thinking	Substitute Holiday
Week 15 6.1.~6.7.	Chemical Separations, IC, CE, FFF, Lab-on-a Chip	6.3.
Core Competencies	♦ problem-solving ability, complex communication, leadership and teamwork, digital and quantitative literacy, science research ability, analytical and creative thinking	Local Election Day
Week 16 6.8.~6.14.	Make-up class week	
Week 17 6.15.~6.19.	Final Exams (6.18.~6.19.)	

※ The above plans may change according to the academic schedule.

※ The schedule will change as it may chance

[Reference] Syllabus Competence Reference

1. Analytical and Creative Thinking	2. Complex Communication	3. Leadership and Teamwork	4. Digital/Quantitative Literacy	5. Global Perspective	6. Adaptability/Progression / Spirit of Adventure	7. Integrity and Ethical Decision Making	8 Habits/Mindset
Ability to identify, manage and deal with complex problems	Ability to understand and express themselves in two or more languages	Ability to be the first to come up with and present new ideas	Ability to understand, use and apply digital technology	Ability to develop an open mind about the traditions or values of others	Ability to develop flexibility, agility and adaptability	Ability to sustain an empathetic and understanding perspective	Sincerity
Ability to see through prejudice and identify reliable information and uncertain information	Ability to communicate clearly with diverse audiences or readers	Ability to influence people to follow themselves, rather than lead them through directives, orders, or authority	Ability to create digital contents and media	Ability to understand the history, politics, religion, and culture of another region	Ability to challenge unfamiliar situations	Ability to cultivate sincerity, honesty, fairness, and respect creativity	Creativity
Ability to find quality information in the flood of information	Ability to listen attentively	Ability to build trust, resolve conflicts, and support others	Ability to utilize multimedia resources to effectively communicate one's ideas in a variety of formats	Ability to use one or more of the international languages used worldwide	Ability to explore and experiment	Ability to display moral courage in the face of injustice	A love of learning and curiosity
Ability to articulate meaningful questions	Ability to speak effectively and coherently	Ability to facilitate group discussions or discussions, build consensus, and negotiate outcomes	Ability to master and use higher mathematics	Ability to use technology to connect people and things around the world	Ability to word effectively in situations of ambiguous or changing priorities	Ability to act responsibly for the benefit and happiness of the greater community	Resilience/elasticity
Ability to analyze and create ideas or knowledge	Ability to write concisely and clearly, considering the diversity of readers	Ability to teach, guide, and counsel Ability to seek help from others	Ability to understand traditional or recently emerging fields such as mathematics, science, technology, environmental science, robotics, geometry, cellular automata, nanotechnology, and biotechnology	Social and intellectual skills to effectively navigate information across multiple cultures	Ability to see failure as a lesson and to accept that innovation is made up of frequent failures and small successes	Ability to have an intrinsic understanding of ethical issues or dilemmas about new media or new technologies	Perseverance/patience
Ability to devise and test solutions to problems by	Ability to explain information and persuasively	Ability to collaborate on tasks, manage groups, and		Ability to use 21 st century core competencies such as communication	Ability to develop a sense of independence to seek new roles,	Ability to make rational and ethical decisions on complex	Self-efficacy

서식 있음: 왼쪽, 단어 잘림 허용

1. Analytical and Creative Thinking	2. Complex Communication	3. Leadership and Teamwork	4. Digital/_ Quantitative Literacy	5. Global Perspective	6. Adaptability/ Progression _{HH} / Spirit of Adventure	7. Integrity and Ethical Decision Making	8 Habits/_ Mindset
trial and error	engage others in an engaging way	delegate responsibilities		skills, collaboration skills, critical thinking skills, creativity to properly understand and deal with global issues	ideas, or strategies	issues	
Ability to think of different ways to cope		Ability to listen and practice goals		Ability to learn from and collaborate with people from different cultures, religions and lifestyle through mutual respect and open dialogue	Ability to develop the knowledge and competencies necessary to start a business and run a business		Stress management skills
Ability to develop knowledge and perspectives that are not limited to one field but span multiple fields		Ability to share credit for success		Ability to generate diverse ideas and exploit social and cultural differences to get things done			Time management skills
Ability to make sustained inferences							
Ability to put things together or adapt them to new uses and situations							
Ability to solve new problems that cannot be solved through established rules							
Ability to use knowledge or creativity to solve complex real-world problems							

서식 있음: 왼쪽, 단어 잘림 허용

1. Complex Problem-Solving Problem-Solving Skills	Ability to solve new and unclear problems that may arise in complex real-world situations
2. Critical Thinking Skills	Ability to compare and analyze the strengths and weaknesses of various solutions or approaches to a given problem using logic and reasoning
3. Creativity	Ability to come up with novel ideas when given a certain topic or situation or to suggest a creative method in the process of solving a problem
4. Human Resource Management Skills	Ability to select the most suitable person for a specific task and manage them to achieve maximum work performance through motivation, <u>and</u> systematic and efficient management
5. Ability to Collaborate	Ability to adjust one's own behavior in response to the behavior of others
6. Emotional Ability	Ability to perceive other people's reactions and understand why they react the way they do
7. Judgement and Decision-making Skills	Ability to make the most rational judgement and decision by analyzing the costs and benefits of a specific situation or problem
8. Orientated Towards Service	Ability to actively seek ways to help others
9. Negotiation Skills	Ability to bring together parties who differ in ideas or interests and try to reduce those differences
10. Cognitive Flexibility	Ability to apply different kinds of rules or principles to rearrange and reorganize things in different ways